INCH-POUND

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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M13 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M13 Cartridge Actuated Thruster Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Indian Head Division, Naval Surface Warfare Center (CAGE Codes 19200 and 19203)

10520974	Thruster, Cartridge Actuated, M13, Metal Parts Assembly
8836128	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M13 or M19
8836129	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M13 or M19
9217388	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M13 or M19

U.S. Air Force

TPO 00-756-3501	Thruster, Cartridge Actuated, M13

(Application for copies should be addressed to: Commander, Naval Surface Warfare Center, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured thruster metal parts assembly shall contain new shear pins, new O-rings, and new setscrews (see 6.5).

3.2 Hydrostatic pressure. The body shall withstand a hydrostatic pressure of 11,500 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.3.1).

3.3 Torque. Torque requirements shall be as specified in table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I. Torque requirements.

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Component	B Seating or Bottoming (inch pounds)	C Breakaway (inch pounds)
Piston and piston fastener	100 ± 20	75 min

3.4 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.3.2) shall have the existing protective finish stripped and the specified protective finish reapplied. When there are only small surface areas of exposed base metal on the aluminum exterior components (body assembly and end cap) it may be permissible to treat the surfaces with chemical film Class 1A as specified in MIL-C-5541.

3.5 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.4.

3.6 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled with sufficient torque to bottom components, except as noted in table I.

3.7.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.7.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for thrusters.

3.8 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual

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examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Ring, Drawing 10520489, covering a detail of Drawing 10520974.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	
Visual		
Minor		
None defined		

4.3.2.3 Pin, Firing, Drawing 10520978, covering a detail of Drawing 10520974.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	
Visual		
Minor		
None defined		

4.3.2.4 Cap, Base, Drawing 10520982, covering a detail of Drawing 10520975.

Critical
None defined

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Major

101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	
	Visual	

Minor
None defined

4.3.2.5 Fastener, Piston, Drawing 10520983, covering a detail of Drawing 10520980.

Critical
None defined

Major

101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	
	Visual	

Minor
None defined

4.3.2.6 Body, Drawing 10520976, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corners	
	Visual	
105	Foreign matter	
	Visual	

Minor
None defined

4.3.2.7 Piston, Drawing 10520984, covering a detail of Drawing 10520979.

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Critical
None defined

Major

101	Threads damaged or not full	Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr or sharp corners	
	Visual	
105	Foreign matter	
	Visual	

Minor
None defined

4.3.2.8 Base Cap and Pellet Assembly, Drawing 10520975, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Pellet missing, loose, or improperly inserted	Manual/Visual
102	Thread damaged	Visual
103	Burr	Visual
104	Foreign matter	
	Visual	

Minor
None defined

4.3.2.9 Piston and Pellet Assembly, Drawing 10520979, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Pellet missing, loose, or improperly inserted	Manual/Visual
102	Thread damaged	Visual
103	Burr	Visual
104	Foreign matter	
	Visual	

Minor
None defined

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4.3.2.10 Piston Fastener and Pellet Assembly, Drawing 10520980, covering a detail of Drawing 10520974.

Critical
None defined

Major

101	Pellet missing, loose, or improperly inserted	Manual/Visual
102	Thread damaged	Visual
103	Burr	Visual
104	Foreign matter	
	Visual	

Minor
None defined

4.3.2.11 Breech and Firing Pin Subassembly, consisting of: Breech, Drawing 10520977; Firing Pin, Drawing 10520978; and O-ring, MS29513-9; covering a detail of Drawing 10520974.

Critical

1	Firing pin protrusion	Gage

Major

101	Protective finish inadequate or defective	Visual
102	Burr	Visual
103	Foreign matter	
	Visual	

Minor
None defined

4.3.2.12 Metal Parts Assembly, Drawing 10520974.

Critical
None defined

Major

101	Piston not free to rotate 360 degrees clockwise	Manual
102	Any part missing, except as otherwise classified	Visual
103	Improper assembly	Visual
104	Thread damaged or not full	Visual
105	Foreign matter	
	Visual	

Minor

201	Protective finish inadequate or defective	Visual

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202	Barrier material missing from setscrews	Visual
203	Protective cap or plug missing	Visual
204	Pellet missing from piston fastener	Visual

4.3.2.13 Breech, Drawing 10520977, covering a detail of Drawing 10520974.

Critical

1	Surface finish or cavity improper	Gage

Major
None defined

Minor
None defined

4.3.2.14 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8836128.

Critical
None defined

Major
None defined

Minor

201	Contents loose	
	Manual	
202	Box damaged	
	Visual	
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.3.2.15 Box, Packing, Ammunition, Drawing 8836129.

Critical
None defined

Major
None defined

Minor

201	Contents loose	
	Visual/Manual	
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	
	Visual	

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205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.16 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217388.

Critical
None defined

Major
None defined

Minor

201	Contents loose	
Manual/Visual		
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loosed, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	
Visual		
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each body shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.2 shall be cause for rejection of the unit.

4.4.2 Breakaway torque. Each piston fastener and piston shall be subjected to a breakaway torque. Failure of any unit to comply with the requirements of column C table I shall be cause for rejection of the unit.

4.4.3 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.4 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.4. Failure of any unit in the inspection sample to comply with 3.5 shall be cause for rejection of the assembly.

4.4.5 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

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4.5.1 Hydrostatic pressure. Prior to application of protective finish the body shall be mounted in a device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

4.5.2 Breakaway torque. The torque wrench set at the torque requirement specified in table I, shall be used in conjunction with a fixture to determine the adequacy of the joint. The torque shall be applied to the piston fastener.

4.5.3 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.5 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8836128 and 8836129. Level B shall be in accordance with Drawings 8836128 and 9217388.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-756-3501.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. The responsible inspection activity (see 4.1)

d. Any special instruction requirements (see 4.2)

e. Level of packing and marking (see 5.1)

6.3 Definitions.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: base cap, body, and piston fastener.

6.4 Old engraved or stamped lot and serial numbers. Old serial numbers and lot numbers engraved or stamped on metal parts shall be over-stamped by x's to obliterate those markings before applications of final protective finish.

6.5 Nylon pellets. Base caps and piston fasteners which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after replating in accordance with the applicable drawing.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

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Air Force - 99

(Project 1377-0E12)

Review activity:
Air Force - 70

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